THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — MATHEMATICS · MID-TERM PRACTICE PAPER

Session 2026-27 | CBSE / RBSE Pattern

Class: 8	Subject: Mathematics	Set 03	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) A perfect square can never end with the digit: **1**

(a) 1 (b) 4 (c) 8 (d) 9

(ii) The cube root of 0.027 is: **1**

(a) 0.03 (b) 0.3 (c) 3 (d) 0.9

(iii) The sum of the first 12 odd natural numbers is: **1**

(a) 24 (b) 144 (c) 132 (d) 156

(iv) The value of $(-3)^{-2}$ is: **1**

(a) 1/9 (b) −1/9 (c) −9 (d) 9

(v) 1999 written in Roman numerals is: **1**

(a) MCMXCIX (b) MDCCCCLXXXXVIIII
(c) MCMXCVIIII (d) MXMIX

(vi) Which of the following is NOT a valid base-4 numeral? **1**

(a) 1302 (b) 2031 (c) 3210 (d) 1234

(vii) Three angles of a quadrilateral are 80°, 100° and 70°. The fourth angle is: **1**

(a) 110° (b) 100° (c) 90° (d) 120°

(viii) The diagonals of a rhombus: **1**

(a) are always equal (b) are parallel
(c) do not intersect (d) bisect each other at right angles

(ix) Which of the following numbers is divisible by 11? **1**

(a) 2728 + 1 = 2729 (b) 3026
(c) 3025 (d) 3035

(x) (x + 2)(x + 3) equals: **1**

(a) $x^2 + 5x + 6$ (b) $x^2 + 6x + 5$ (c) $x^2 + 5x + 5$ (d) $x^2 + 6$

(xi) If $a + b = 10$ and $ab = 21$, then $(a - b)^2$ equals: **1**

(a) 79 (b) 58 (c) 100 (d) 16

(xii) A car travels 150 km in 3 hours. At the same speed it will travel in 5 hours: **1**

(a) 200 km (b) 300 km (c) 250 km (d) 350 km

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the cube root of 1728 by prime factorisation. **2**

3. Write 3.56×10^5 and 2.1×10^{-3} in usual form. **2**

4. Write 1999 in Roman numerals. **2**

5. Add $(1101)_2$ and $(1011)_2$ and give the answer in base 2. **2**

OR

Convert the decimal number 100 into base 7.

6. In parallelogram ABCD, $\angle A = (2x + 10)°$ and $\angle C = (3x - 20)°$. Find x and $\angle A$. **2**

7. Find the ratio of 3 m 60 cm to 2.4 m in simplest form. **2**

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Without actual division, check whether 678432 is divisible by (i) 8 (ii) 11. Give reasons. **3**

9. In parallelogram ABCD, AB = (2x + 3) cm and DC = (3x − 5) cm. Find x and the length of AB. **3**

OR

The diagonals PR and QS of a rectangle PQRS meet at O. If $\angle POQ = 110°$, find $\angle OPQ$.

10. Using suitable identities, find: (i) 97×103 (ii) 205^2 (iii) 99^2 **3**

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 3528 must be divided so that the quotient is a perfect square. Also find the square root of the quotient. **5**
(ii) Find the square root of 0.0289.

OR

(i) Find the smallest number by which 1125 must be multiplied so that the product is a perfect cube. Also find the cube root of the product.
(ii) The volume of a cubical tank is 0.343 m^3. Find the length of its edge.

12. Write in standard form: (i) 4,56,00,000 (ii) 0.00000072. Write in usual form: (iii) 7.3×10^{-4} (iv) 5.2×10^6. **5**
(v) Find $2 \times 10^5 + 3 \times 10^4$ in standard form.

OR

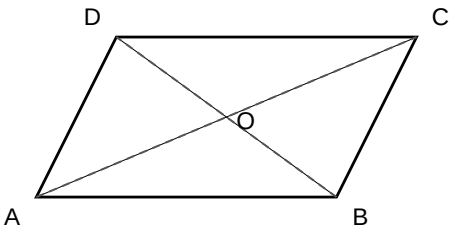
Simplify: (i) $(2^3)^{-2} \times 2^7$ (ii) $6^5 \div 6^{-2}$ (iii) $(4^{-1} + 2^{-1})^{-1}$ (iv) $(10^{-3})^{-2}$ (v) $(a^2b)^3 \div (ab)^2$

13. (i) Write the ratio of 2.5 kg to 750 g in simplest form. **5**
(ii) Divide ₹ 1,800 among A, B and C in the ratio 2 : 3 : 4.
(iii) If 8 pens cost ₹ 96, find the cost of 15 pens.
(iv) Find x if 6 : x :: 9 : 15.
(v) The ratio of boys to girls in a class is 5 : 4 and there are 36 students. Find the number of girls.

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A flooring contractor cuts parallelogram-shaped tiles ABCD. In one tile, AB = 12 cm, BC = 8 cm and ∠A = 65°. The diagonals AC and BD intersect at O.



(i) Write the length of CD. 1

(ii) Find ∠B. 1

(iii) Show that AO = OC. 2

OR

(iii) Find the perimeter of the tile and the cost of an edging strip around 50 such tiles at ₹ 2 per cm.

15. Kabir arranges square tiles to form bigger and bigger squares: 1 tile (1×1), 4 tiles (2×2), 9 tiles (3×3), and so on. He notices that each new square needs the next odd number of extra tiles.

(i) How many tiles are needed for the 8th square? 1

(ii) How many extra tiles are needed to change an 8 × 8 square into a 9 × 9 square? 1

(iii) Show that the sum of the first 10 odd numbers is a perfect square and name it. 2

OR

(iii) A square arrangement uses 441 tiles. How many more tiles are needed to make the next larger square?

16. Three friends — Ria, Sana and Tej — travel by a shared taxi for different distances: Ria 3 km, Sana 5 km and Tej 7 km. The total fare of ₹ 450 is divided in the ratio of the distances travelled.

(i) Write the ratio of the shares of Ria, Sana and Tej. 1

(ii) Find Sana's share of the fare. 1

(iii) Find the difference between Tej's share and Ria's share. 2

OR

(iii) Another day the same three friends pay ₹ 600 for the same distances. Find the amount Ria pays and show that the ratio 3 : 5 : 7 is unchanged.